Introduction

The **ProISP IV** from **PROCHILD** is a low cost In-System Programmer covering all **AVR MCUs** with **ISP, PDI** or **TPI** Interface.

<Features>

- Stand-alone **AVR** Programmer(No PC needs)
- Easy to use **GUI**
- Programs both **Flash** and **EEPROM**
- Supports **Fuse** and **Lock** byte programming
- Supports **RC Oscillator Calibration byte** programming
- Supports **Serializing**
- Upgradable for future devices
- **USB** Interface

<Contents>



ProISP IV



USB cable

Getting Started

1. Install 'ProISP IV PROGRAMMER' and USB driver

- **USB driver**

  Before connecting ProISP IV to your PC you have to install a USB to UART Bridge VCP(CP2102) driver

- **Software**

  In order to use the **ProISP IV** it is required to install the **ProISP IV PROGRAMMER**. You can download on our website(see below)

<Download>

- **http://www.prochild.co.kr -> DATA room**
  - **-> S/W : ProISP IV PROGRAMMER Version x.xx**
  - **-> Driver : [Silicon Labs]CP210x USB to UART Bridge VCP Driver**
    (It can be downloaded from http://www.silabs.com)

2. Mode Switch

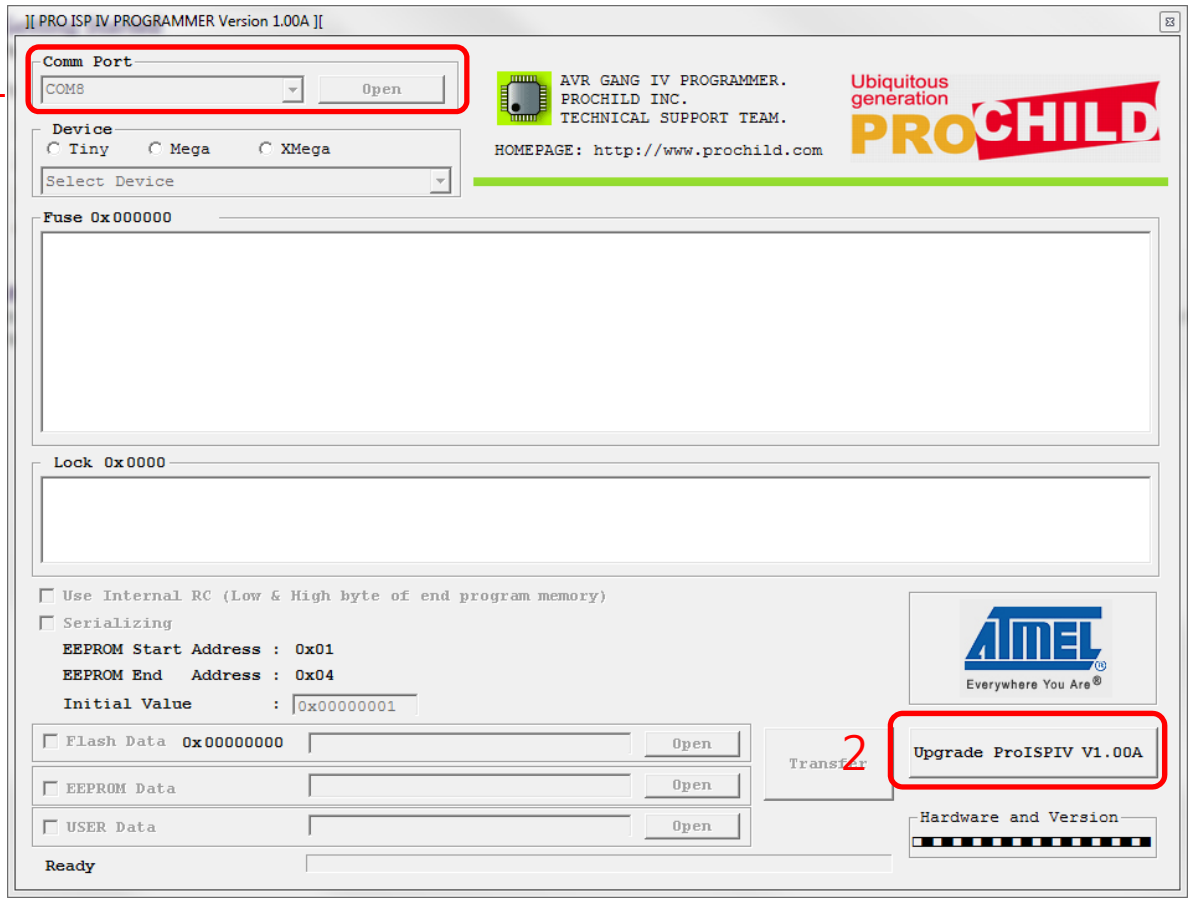
- Set the **Mode Switch** to '**U**' when you upgrade ProISP IV or update source(**.hex)** code
- After update code, you should set the **Mode Switch** to '**T**' for target writing

Getting Started(Cont.)

3. Upgrade **ProISP IV**

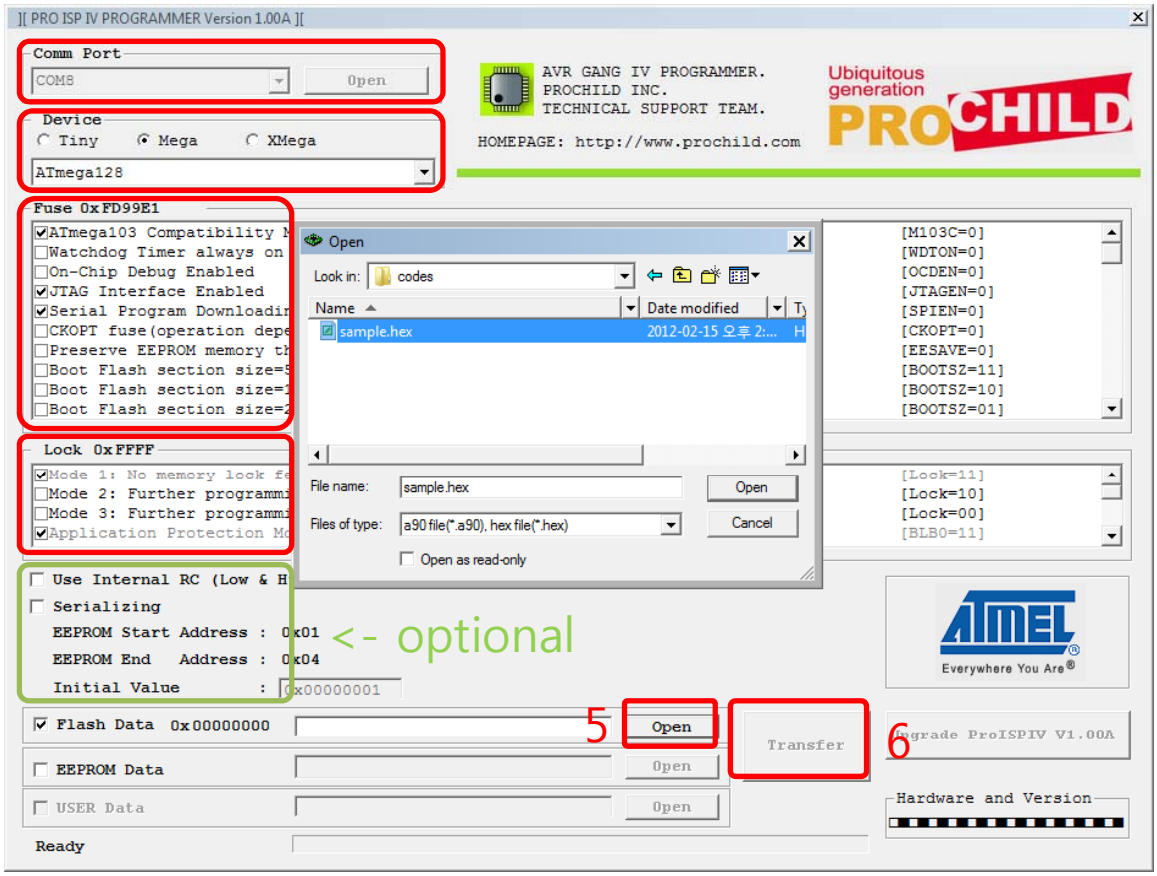
- Start '**ProISP IV PRGRAMMER'.**
- Connect **ProISP IV** to PC using USB cable.
- Set the **Mode Switch** to **'U'**
- Open a correct **Com Port.**

  (Check it on the *Device Manger* provided by windows.)



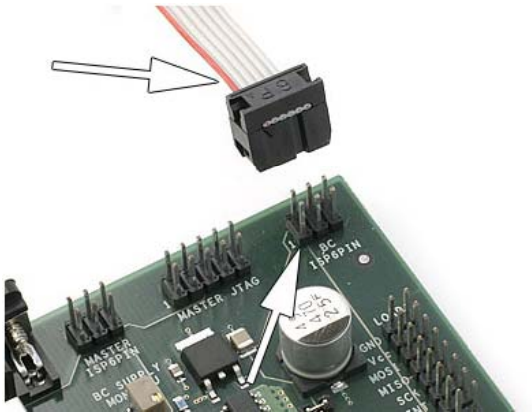
Download & Target Writing

1. To program your target devices, you have to download your code to the **ProISP IV** first.
   - Open **Com Port** → Select **Device** → Set **Fuse** → Set **Lock** → Open a code(.**Hex**) → **Transfer**

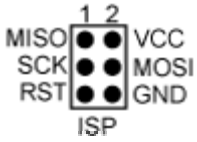


Download & Target Writing

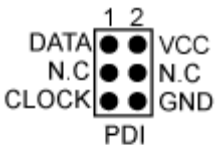
2. Connect ProISP IV to target board.



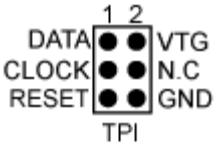
ISP Interface



PDI Interface



TPI Interface



(Programming voltage for **TPI** is **5V** only)

3. Push the Programming Button.

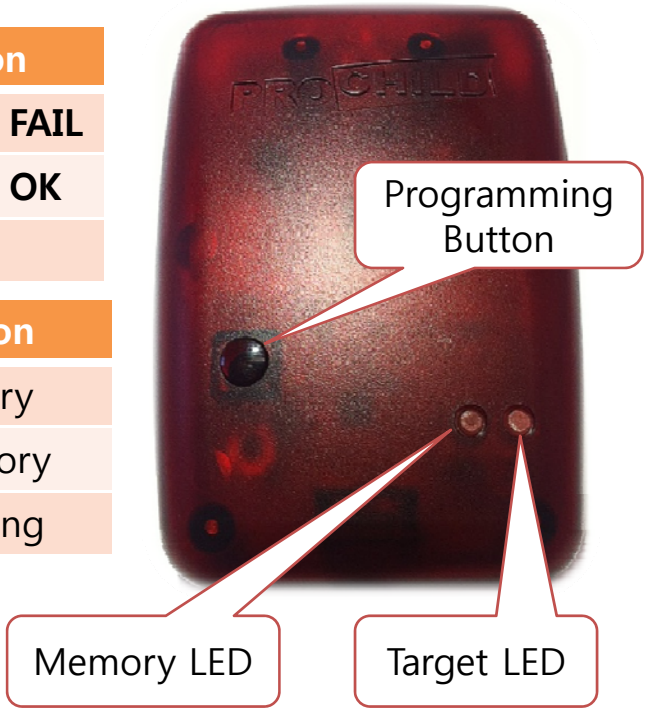
(The power for **ProISP IV** must come from **Target board**)

Hardware Description

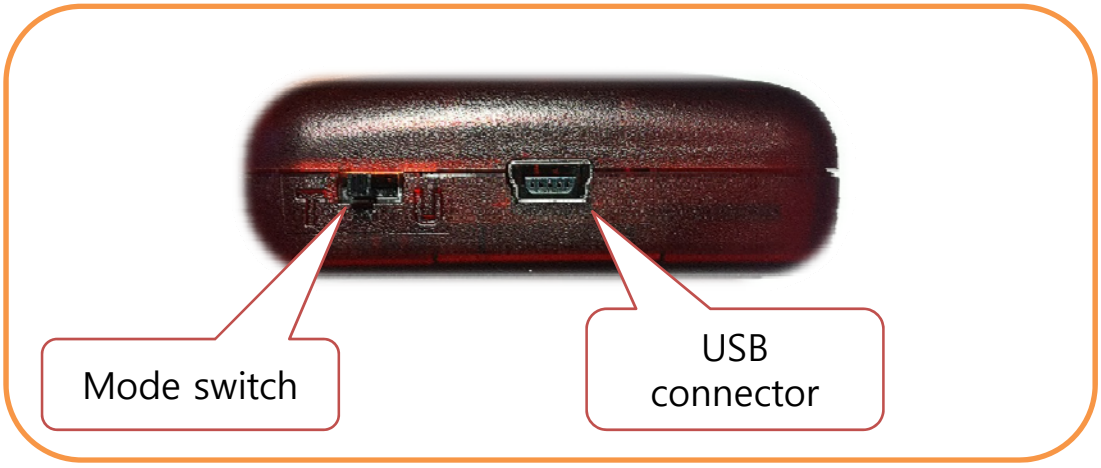
1. Status LEDs

Target LED	Description
Red	Programming **FAIL**
Green	Programming **OK**
Orange blinking	Target writing

Memory LED	Description
Red	Empty memory
Green	Loaded memory
Orange blinking	Memory writing



2. Rear panel



Int. RC calibration bytes

- By changing the **OSCCAL** register from S/W, it is possible to get a higher calibration accuracy than by using the factory calibration.

- **ProISP IV** is designed to assign the user calibration value to the last two bytes of Flash memory. Therefore, the location of the data is determined by a Flash memory size.(**See below**)

<Example>

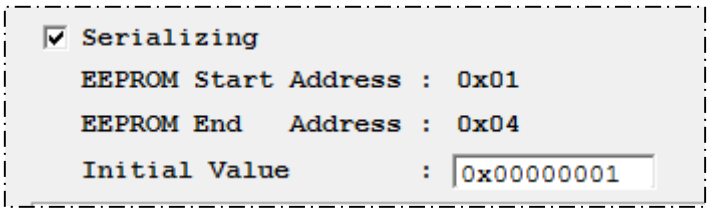
- Device : ATmega16, 16KB flash memory
- Address : 0x3FFE, 0x3FFF
- Example source code

```
//Write user calibration bytes to OSCCAL
asm("LDI    R30,    0xff"); //address of user data(0x3fff)
asm("LDI    R31,    0x3F");
asm("LPM                 ");
asm("OUT    0x31,   R0   "); //address of OSCCAL(0x31)
```

Serializing

- **ProISP IV** can write a user **serial number** on the **internal EEPROM** of each target devices.

<How to use the **Serializing**>



- Check **Serializing.**
- The address of the serial number is **0x0001~0x0004.** (**The address can not be changed.**)
- **Initial Value** is the first serial number of the target device.
- Whenever you write a target device, the **serial number** is increased by 1 automatically.